

Semyon Fox

Galway, Ireland – available on-site in San Francisco, Jan–Aug 2027 · [✉ semyon.fox@gmail.com](mailto:semyon.fox@gmail.com)
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Spring/Summer 2027 Engineering Intern

Third-year Computer Science and IT student who builds practical product software for learning and sports communities, from React and Next.js interfaces through Node.js services and PostgreSQL data models to deployment and operations.

Technical skills

Languages: TypeScript, SQL, Python, Go, Rust, Bash

Product engineering: React, Next.js, Node.js/Express, REST APIs, PostgreSQL, Redis, Qdrant

AI & integrations: Retrieval-augmented chat, Model Context Protocol, Canvas LMS API, BullMQ job queues

Delivery: Docker, Jenkins CI/CD, AWS (Fargate, RDS, S3, ElastiCache), Linux, Nginx, Cloudflare

Education

University of Galway

BSc (Honours) Computer Science and Information Technology

Expected Aug 2028

Galway, Ireland

Third year; current overall average: 2:1.

Relevant coursework: Software Engineering; Web Development; Database Systems; Human Computer Interaction; Data Structures and Algorithms.

Projects

OghmaNotes – AI study platform

Next.js, TypeScript, PostgreSQL, Qdrant, Redis/BullMQ, Docker

Deployed

oghmanotes.ie

- Built in a three-person team: Markdown notes, RAG chat, Canvas import, quiz generation, and FSRS spaced repetition.
- Built an asynchronous Canvas import pipeline with PDF extraction, embeddings, BullMQ queues, live job status, retries, and cancellation.
- Implemented semantic search and retrieval-augmented chat using Qdrant for vectors and PostgreSQL for content and ownership metadata.

Canvas MCP – API integration

TypeScript, Model Context Protocol SDK, Zod

Open source

github.com/semyonfox/canvas-mcp

- Consolidated and extended community tool designs into a typed MCP server exposing selected Canvas LMS APIs to assistants.
- Added validated inputs, pagination, request timeouts, and bounded retries for read operations.

Uisce – swim club management

React, Express, PostgreSQL, Redis, JSON:API, Docker, Vitest

Deployed

swim.semyon.ie

- Built club-scoped workflows for coaches and athletes: onboarding, roles, squads, sessions, attendance, results, and analytics.
- Designed a multi-schema PostgreSQL database for authentication, training, and competition data, with a JSON:API backend.

Home lab and CI/CD infrastructure

Linux, Docker, Jenkins, Nginx, Cloudflare, Btrfs, NFS

Self-hosted

- Improved Jenkins CI/CD pipelines with persistent BuildKit caching, reusing Docker build layers across runs.

Collaboration & leadership

University of Galway Computer Society

Treasurer; previously Auditor and PR lead

Nov 2024–Present

Galway, Ireland

- Organised CTF 2026 for 110+ participants with four corporate sponsors, coordinating technical and non-technical work.
- Contributed merged website changes covering build/deployment fixes, routing and committee content; also contributes fixes and improvements to open source through pull requests.

Interests

Irish language and culture, swimming, filmmaking, and student learning tools.